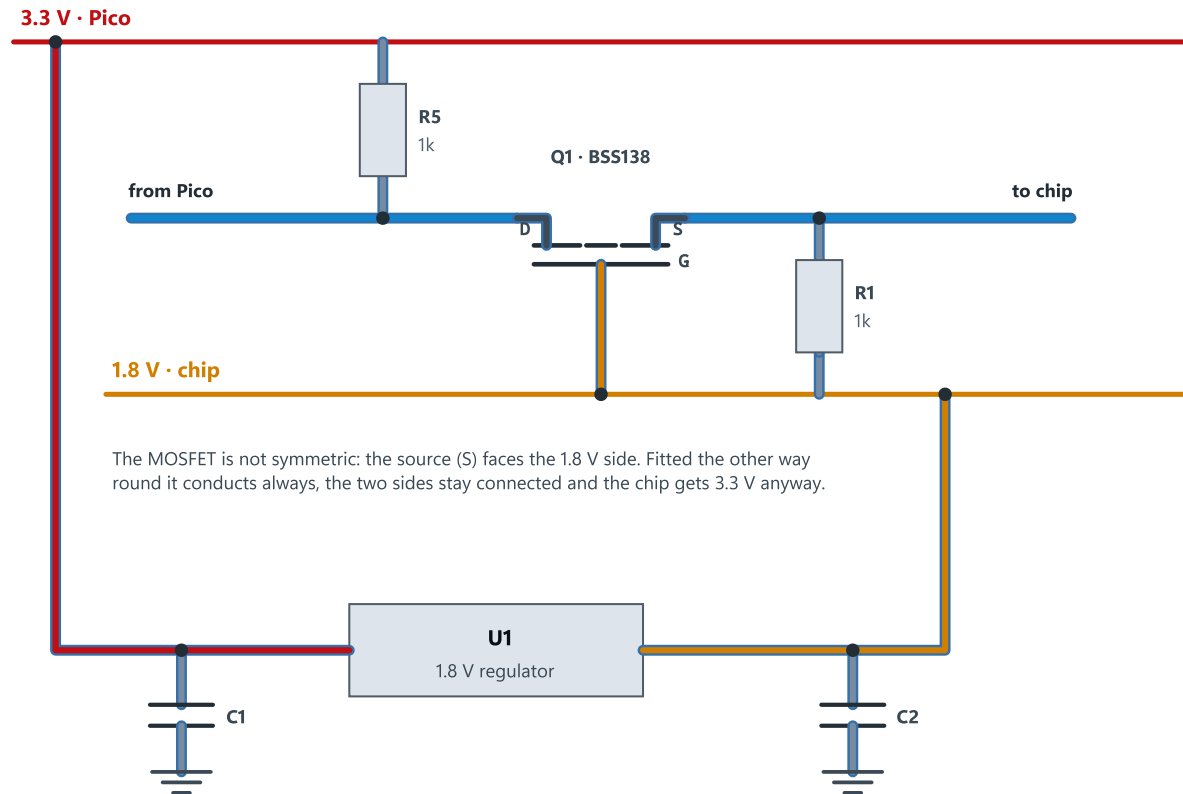


Level shifter for 1.8 V chips

One of four identical channels · BSS138 · 3.3 → 1.8 V regulator · shared ground



The two capacitors are not decoration: without them the regulator oscillates and the chip supply wanders.

Level shifter for 1.8 V chips

Bill of materials

Ref	What	Part numbers
Q1-Q4	N-MOSFET · $V_{gs(th)} < 1.5\text{ V}$ · SOT-23	onsemi BSS138LT1G · Diodes BSS138-7-F · Nexperia BSS138BK
R1-R8	1 k Ω · 1% · 0603	Yageo RC0603FR-071KL
U1	LDO 1.8 V · $> 100\text{ mA}$ · SOT-23	Microchip MCP1700T-1802E/TT · Torex XC6206P182MR · Diodes AP2112K-1.8TRG1
C1, C2	1 μF · X7R · 16 V · 0603	Murata GRM188R71C105KA12D

- Off the shelf: SparkFun BOB-12009 or Adafruit 757 (four BSS138 channels), plus the 1.8 V regulator separately.

One channel per signal

Signal	Pico		Chip
SCLK	GP2	→	6 CLK
MOSI	GP3	→	5 DI
MISO	GP4	←	2 DO
CS	GP5	→	1 /CS

Worth knowing

- The chip takes 1.8 V from the regulator, NOT the Pico's 3V3. Translating the signals and leaving the supply at 3.3 V is the quickest way to destroy it.
- The rising edge is made by the resistor, not the transistor. 1 k Ω holds 4 MHz; the textbook 10 k Ω (which come from 100 kHz I²C) give a 700 ns rise, and the two reads already disagree at 1 MHz.
- On the MOSFET what matters is the threshold, not the current: the gate sits at 1.8 V, so $V_{gs(th)}$ must be under 1.5 V. The BSS138 has it, the 2N7002 does not (up to 2.5 V) and never turns on — same package, same price, does not work.
- To keep 12 MHz you need a fixed-direction translator, TI SN74LVC8T245PWR. Not the TXS0108E: it is made for open-drain buses and does not work on SPI.